$$\mu_{\text{cam}} = W\mu + t, \quad \bar{u} = \pi(\mu_{\text{cam}}),$$
$$\Sigma_{\text{screen}} = JW\Sigma W^{\mathsf{T}} J^{\mathsf{T}},$$
$$W \in \mathbb{R}^{3\times 3}, \quad J \in \mathbb{R}^{2\times 3}$$